Power Budget - LiDAR Subsystem (Daughter Board)

Team Number: Team 1 (Group 101)
Project Name: **OpenField**
Team Member Names: **Landon Kuhman (LiDAR subsystem)**
Version: **1.2 (10/28/2026)**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ToF LiDAR (D500)	STL-19P	+4.5 - 5.5V	1	290	290	mA
	PIC18F57Q43 Nano	DM164150	+4.4 - 5.25V	1	100	100	mA

+12V Rail (unused)	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None			0	0	0	mA
							mA
							mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
Regulator	Not used	N/A	N/A	0	0	0	mA
					Total Remaining Current Available on +12V Rail	0	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ToF LiDAR (D500)	STL-19P	+4.5 - 5.5V	1	290	290	mA
	PIC18F57Q43 Nano	DM164150	+4.4 - 5.25V	1	100	100	mA
	Indicator LEDs (D1 power, D3/D4 debug)	Green/red LEDs, 330 R / 1 k	+3.3 - 5V	3	4	12	mA
					Subtotal	402	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	502.5	mA
Regulator	+5V LDO Regulator	LM2940CT-5.0	+6V - 26V	1	1000	1000	mA
					Total Remaining Current Available on +5V Rail	497.5	mA

+3.3V Rail (unused)	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None			0	0	0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	0	mA
Regulator	Not used	N/A	N/A	0	0	0	mA
					Total Remaining Current Available on 3.3V Rail	0	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	3S Li-ion pack (3 x 18650 in series)	POWOBEST POWO18B 3000 mAh (x3)	+9.0 - 12.6V	11.1V nom.	3000	3000	mA

Power Rails Connected to External Power Source 1	+5V rail (LM2940CT-5.0 input)	LM2940CT-5.0	+6V - 26V	1	502.5	502.5	mA
					Total Remaining Current Available on External Power Source 1	2497.5	mA

External Power Source 2 (bench)	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 2 Selection	9 V DC wall adapter	2.1 mm barrel plug (fits J1, PJ-063AH)	100 - 240V AC	9V DC	1000	1000	mA
Power Rails Connected to External Power Source 2	+5V rail (LM2940CT-5.0 input)	LM2940CT-5.0	+6V - 26V	1	502.5	502.5	mA
					Total Remaining Current Available on External Power Source 2	497.5	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	3S Li-ion pack (3 x 18650 in series)	POWOBEST POWO18B 3000 mAh (x3)	+9.0 - 12.6V		3000	502.5	
						Battery Life	6.0	hours

Notes:

On the robot, the board runs from a 3S pack of 3.7 V 18650 cells in series (9.0 V empty, 11.1 V nominal, 12.6 V full), plugged into J1. On the bench, a 9 V adapter plugs into J1 instead. Only one source is connected at a time. A linear regulator draws about the same current from its input as it delivers, so each source supplies about 502.5 mA. Battery max current assumes 1C (3000 mA) continuous because no cell datasheet is available.

Regulator heat is (Vin - 5 V) x I: about 2.0 W at 9 V and about 3.8 W at 12.6 V (full pack), so U2 needs a TO-220 heatsink. Cells are charged one at a time with TP4056 modules, removed from the series pack.

Curiosity Nano current (100 mA) is a conservative allowance for the PIC18F57Q43 at 64 MHz plus the on-board nEDBG debugger; D500 current (290 mA) is the rated working current (measured 247 mA on the bench). LEDs are about 4 mA each.